

Florida Hurricane Catastrophe Model

Informal Internship Execution Plan

Working checklist

June 2026

1 End goal

By the final presentation, have one clear story:

I built a transparent mini catastrophe model. It maps storm-track hazard and property vulnerability into policy-level insured loss, then aggregates losses to portfolio risk metrics. It is not a vendor replacement, but it explains the main mechanics and gives a reproducible scoring pipeline for hypothetical storms.

Do not try to impress by overengineering. Impress by being structured, reproducible, and honest about limitations.

2 Reading order

1. Re-read Grossi and TeHennepe first. Extract only the model architecture and output metrics: event, hazard, vulnerability, financial module; AAL, EP curve, return periods, CV.
2. Re-read Grant and Atkinson second. Extract only the practical lessons: percent loss target, building variables matter, geography/wind-field missingness explains weak fit, hip roof/newer build signs.
3. Read Powell/Florida public model material only enough to understand local wind-field logic. Do not attempt to reproduce the full wind model.
4. Read Pinelli/Pita only enough to justify a hybrid vulnerability model.
5. Read Landsea/Franklin only for limitations: track and intensity data have uncertainty.

3 Step-by-step plan

Step 1 - Set up clean project structure

Create folders:

- data/raw
- data/processed
- src
- outputs/tables
- outputs/figures
- presentation

Keep raw CSVs untouched. Every processed file should be generated by code.

Step 2 - Basic data audit

For each CSV, produce a small audit table:

- row count and column count;
- missingness by column;

- distinct `hurr_id` count;
- distinct property count;
- numeric min/median/mean/max;
- obvious impossible values: negative loss, zero replacement value, invalid latitude/longitude, impossible year built.

Output: `data_audit.csv` and 3-4 simple charts.

Step 3 - Build the modelling table

Left join claims onto exposure using `hurr_id`, `property_id`. Missing claims become zero loss.

Create:

- `loss` = 0 if missing;
- `claim_flag` = 1 if `loss` > 0;
- `loss_ratio` = `loss` / total replacement value;
- `deductible_ratio` = deductible / total replacement value;
- `age` = hurricane season - year built;
- `log_trv` and `log_gross_area`.

Check whether claims are already insured paid losses. If yes, do not subtract deductible again when training.

Step 4 - EDA that actually matters

Do not waste time on decorative plots. Produce these:

- total loss by hurricane;
- claim frequency by hurricane;
- loss ratio distribution, including positive-only distribution;
- loss ratio by construction, occupancy, age bucket, stories;
- map/scatter of exposed properties and claims;
- storm tracks over property locations if possible.

The aim is to know whether the data are usable and whether losses are concentrated in a few storms/regions.

Step 5 - Engineer hazard features

For each exposure row, use all track snapshots for that hurricane.

Start simple:

- minimum distance from property to storm centre;
- maximum reported storm wind during relevant snapshots;
- wind at closest approach;
- pressure at closest approach;
- Saffir-Simpson category at closest approach;
- landfall distance at closest approach;
- right-of-track indicator;
- rough local wind proxy using wind speed, distance, and quadrant radius.

Do not get stuck building a perfect wind model. The deliverable needs a defensible hazard proxy, not a meteorology PhD.

Step 6 - Build baseline models

Fit these in order:

1. Exposure-only claim occurrence model.
2. Exposure-only severity model on positive claims.
3. Hazard-only claim occurrence model.
4. Hazard plus exposure claim occurrence model.
5. Hazard plus exposure severity model.

This sequence proves whether hazard features add value. That is likely the central result.

Step 7 - Main model

Use a two-part model:

- logistic regression or regularised logistic regression for claim probability;
- lognormal severity model for positive loss ratio;
- expected policy loss = claim probability $\times$ conditional severity $\times$ replacement value.

Keep the feature set small. Add only interactions you can explain: local wind with construction, local wind with age, local wind with stories, and local wind with deductible ratio.

Step 8 - Validation

Use leave-one-hurricane-out validation. Avoid random train/test split because it leaks event information.

Report:

- occurrence AUC/log-loss;
- positive-severity MAE/RMSE;
- actual vs predicted event loss by hurricane;
- calibration by predicted-risk bucket;
- top 10 worst prediction misses and likely reason.

Step 9 - Portfolio outputs

Aggregate predicted losses by hurricane:

- event loss table;
- AAL;
- CV;
- EP curve;
- 1-in-50, 1-in-100, 1-in-250 return-period loss if event probabilities are available or a synthetic catalogue assumption is allowed.

If event probabilities are not given, state clearly that the historical-event output is an empirical scenario set, not a full stochastic catalogue.

Step 10 - Robustness checks

Run only useful robustness checks:

- exposure-only versus hazard plus exposure;
- with and without extreme loss-ratio cap;
- alternative severity model: lognormal versus Gamma;
- different wind-distance decay parameters;

- with and without geography fixed effects.

The point is not to find the highest metric. The point is to show whether conclusions survive reasonable modelling choices.

Step 11 - Scoring pipeline

Create a clean function or script:

Input: exposure table + hurricane track table.

Output: policy-level expected loss, confidence interval, and event-level portfolio loss.

The scoring pipeline is more valuable than a notebook full of one-off cells.

Step 12 - Presentation deck

Start the deck before the model is final. Suggested structure:

1. Problem: why hurricane cat modelling matters for ILS.
2. Data: claims, exposure, hurricane tracks.
3. Architecture: hazard, vulnerability, financial module.
4. EDA: where losses occur and what drives them.
5. Model: two-part hybrid model.
6. Validation: policy-level and event-level performance.
7. Portfolio outputs: AAL, EP curve, tail metrics.
8. Robustness and limitations.
9. Recommendation: useful educational model; not production vendor replacement.

4 Suggested timeline

Time	Work
Days 1-2	Data audit, joins, missingness, obvious data issues.
Days 3-4	EDA and first exposure-only baseline.
Days 5-7	Hazard feature engineering from track data.
Days 8-10	Two-part model and leave-one-hurricane-out validation.
Days 11-12	Portfolio aggregation: AAL, CV, EP curve, return-period framing.
Days 13-14	Robustness checks and uncertainty intervals.
Days 15-16	Clean scoring pipeline and code refactor.
Days 17-18	Build final deck and tables.
Final days	Rehearse explanation, fix weak points, prepare Q&A.

5 Likely final recommendation

A good final answer will probably say:

The model is suitable as a transparent internal learning and diagnostic tool. It reproduces the core catastrophe modelling logic and identifies the importance of storm-track hazard features. It is not suitable as a standalone production pricing model because the event catalogue is limited, wind-field reconstruction is approximate, and claims/exposure data may omit important modifiers such as roof details, terrain, and water intrusion.

That conclusion is not weak. It is exactly the kind of honest conclusion that is defensible.